

Building Classification From Aerial Imagery

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Abstract—This document is a template demonstrating the Ieee-conference format.

Index Terms—template; demo

I. INTRODUCTION

Travel forecasting models are based heavily on data provided by government statistical agencies, such as the US Census Bureau and the Bureau of Labor Statistics. These agencies provide the land use data needed to construct these models, giving statistics such as household number, median household income, etc. for census areas. In countries where statistical agencies are not as reliable, travel forecasting models are generally not developed or have data gaps. In countries where they are available, these models are highly reliant on the public release of such statistical data, which leads to problems if such survies are delayed or canceled.

Recent research has sought to find alternative methods to collect land use characteristics. Gathering, analysing, and utilizing remote sensing data from satellites has been a major focus over the last several decades. Satellite aerial imagery has been used to analyze the volume and density man-made structures and roads to estimate area populations, in both urban and rural settings [Allaw et al. [1]][2]. Day time imagery has also been used to track vehicle movement between major cities as an indication of economic activity [3]. It has also been used to classify land use by estimating landscape patterns and building functions, dividing different surfaces, and then categorizing building types [4]. Night time light data has also been shown to be useful as well. One study correlated city light output with social economic variables such as population, GDP and electric power consumption [5], while another used the light emitted from cars at night on highways to predict vehicle travel between cities [6]. Satellite imagery has also been used to produce data sets such as the Microsoft Building Footprints data set, which has been used along with LiDAR and point of interest data to inventory residential structures from census data [Microsoft [7]][8].

With the rise of commercially available and affordable aerial drones, the ability to capture remote sensing data such as high quality aerial imagery has become significantly easier. The

creation of open source softwares for drone use has also made it more accessible for use in research [9].

Though the practical uses, economical feasibility, and ethical concerns of collecting the higher resolution data is still being discussed [10].

More recently, with the advancement of more accessible deep learning models, machine learning has become very prevalent in analysing and classifying remote sensing data. One of its primary uses so far has been to analyze large amounts of data, for such uses of tracking vehicle movement between cities [3], or predicting population characteristics and locations of man-made structures across entire countries. [2] Building classification is of particular interest for collecting land use characteristics.

“You Only Look Once” (YOLO) being used to identify buildings and extract building footprints. [11]

[12] - Extracting building footprints from point cloud data. Buildings were classified before extracting the footprints. Trains an AI model to classify

This leads to the question “can remote sensing data be used to develop land use data to supplement or replace data from statistical agencies?” In this paper the process, and the results of the trained model will be compared to other cities in Utah.

This paper presents a model that recognizes and categorizes building types based off remote sensing data and estimates household statistics for a general area.

II. METHODOLOGY

A. Data Collection

The city of Logan, Utah was chosen for analysis. Building footprints for Logan were obtained from the Microsoft Building Footprint public dataset. This dataset allowed for easy access to building footprint data for this initial study. All analysis was run using Python 3.12 and various python packages and APIs.

With the chosen city the latitude and longitude of the official city boundaries was obtained. Using this boundary, all the building footprints in the city were extracted from the Microsoft Building Footprint dataset and saved as a geojson file. The buildings were then classified using a simple K-mean

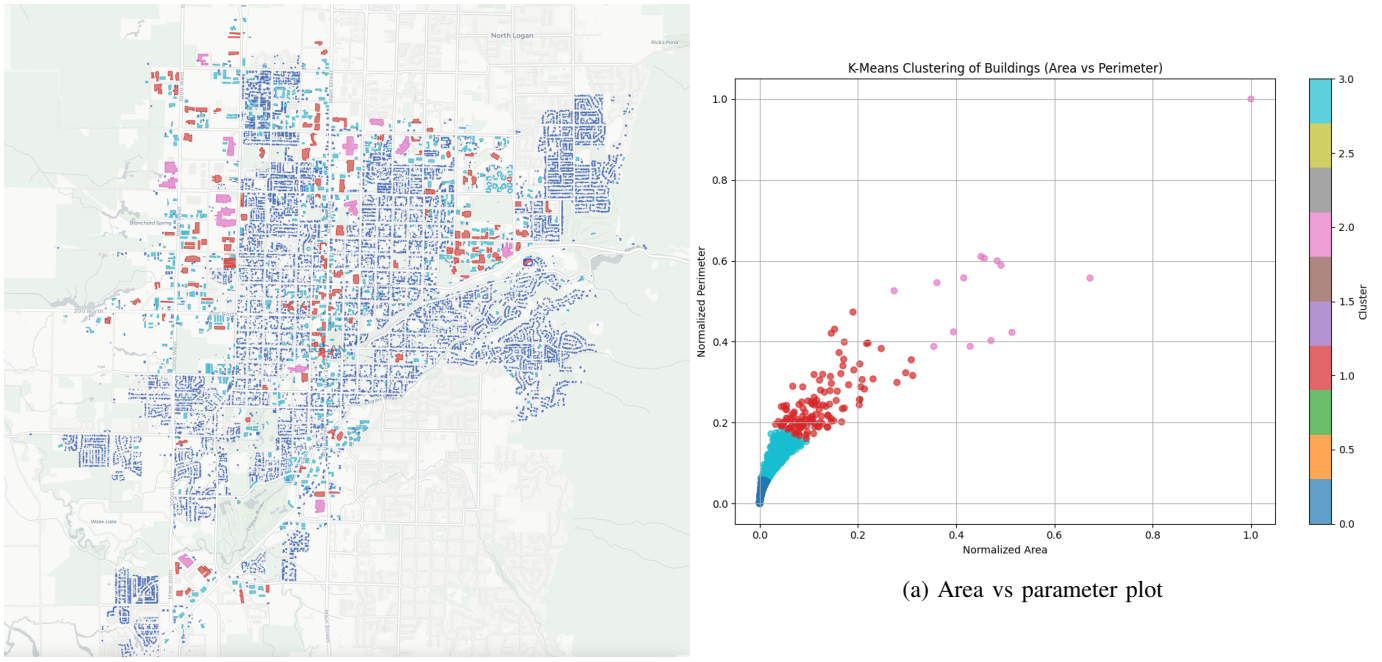


Figure 1: K-mean analysis for Logan, Utah

analysis to differentiate different sized buildings. To do this, the optimal K-mean was found using the elbow method. Using this K value, a K-mean analysis was run comparing the areas of the building footprints to their perimeters. Figure 1 shows the distribution of different cluster types, with most of the buildings in Logan being a smaller size.

Using the building footprint location and the python Library “Contextily”, 600 randomly chosen aerial images of building in Logan were taken from publicly available satellite imagery and cropped to focus on a single building. 200 of the images were from clusters that included only larger buildings to help balance building types in the dataset.

B. Model Training

Ultralytics’ “You Only Look Once” or YOLO model is a open source object detection model that has become very popular within do due it’s accuracy and ease of use. The model is trained on images that have been previously annotated by hand to identify the categories desired for the model to detect. The model trains itself on these images over many iterations until it is able to consistanly recongnize the different categories of objects. The YOLO model is then able to predict these categories other images using bounding boxes and probabilities of correct identification.

Using the open source tool Label Studio, the images were mannually annotated. Every building in each picture was label into five building categories: Apartments, Residential, Commercial, Church, and School.

The annotated images were split into two categories. 80% of the pictures were used in training the object dection model, while 20% of the images were reserved for validation purposes. Within these image sets clusters that included larger

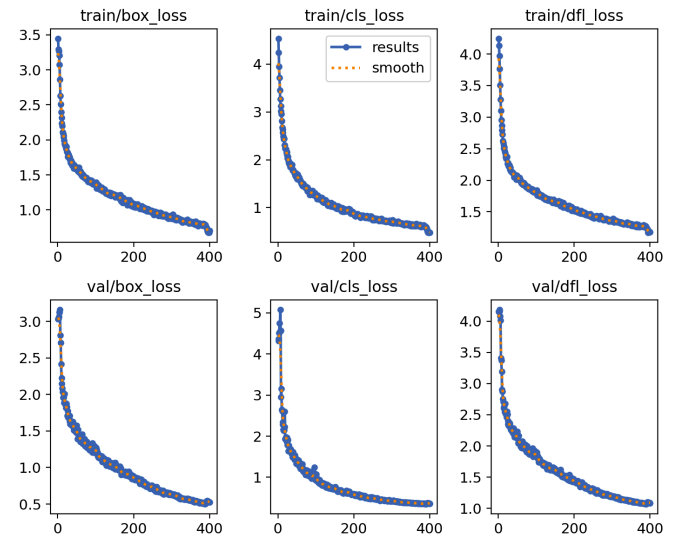


Figure 2: Model training results

and less common building sizes were specifically included, due to thier relatively lower likelihood of appearing, to make sure the model would be trained on these less common building types.

The model was run through hundreds of iterations until the model was able to Figure 2

C. Model Validation and Comparison

Using trained model to predict building types in Logan Utah

Choosing cities in Utah using the American Community Survey data. Based off median income level and age of homes.

Using the model to predict building types in other cities in utah based.

III. RESULTS AND DISCUSSION

The Logan trained model was able to accurately predict residential, commercial, and apartment building types almost 100% of the time, as shown in Figure 4. The desired outcome is to have all the numbers in a diagonal line, which would indicate that the model is predicting the category of a building that is in line with the annotations made.

The model did underperform with schools and churches. This is attributed to the fact that there were very low number of images of both building types provided during model training.

A important aspect to discuss is the model predicting different building types where there is no building. This is seen on the far right of each graph, where the background was categorized. This error occurs from two sources. One is the annotation. Building such as backyard sheds were not identified, but the model still picked them up. Or there were times that only a fourth of a building was present in an image, and so was not annotated, but was still recognized by the model at a lower probability level. The other source of error is the quality of the photos used. The satellite images from Contextily were convenient and easy to access, but were not high quality. This resulted in the model sometimes interpreting things of solid colors, such as roads and covered parking as a type of building. The second error could be solved using higher quality photos, while the first

Confusion matrix for other analysis cities. What is the results? Was there a difference between how the model did between the cities? Was age of the house a big factor? Was income? Is there something else? Neither house age or house median income made a difference in the classification of houses.

Will it be easier to develop a model to be used at different cities, or a process for people to develop their own models?

Talk about how this data is still subject to many of the problems the survey data still is. However, much of this research is susceptible to similar problems as transportation models. Satellite remote sensing data is not consistent around the world,

IV. CONCLUSION AND FUTURE RESEARCH

The conclusion goes here.

As discussed by Hall et al., there are potential ethical problems with such data collection. It straddles the line between public and private information. This will need to be considered as further research is done into this area as to stay on the right side of the law.[10]

What does the future research look like?

Training models to detect other things such as cars, swing sets, patios, etc.

Creating the process to predict household types based off the collected data

Incorporating drone aerial imagery and extracting building footprints from that.

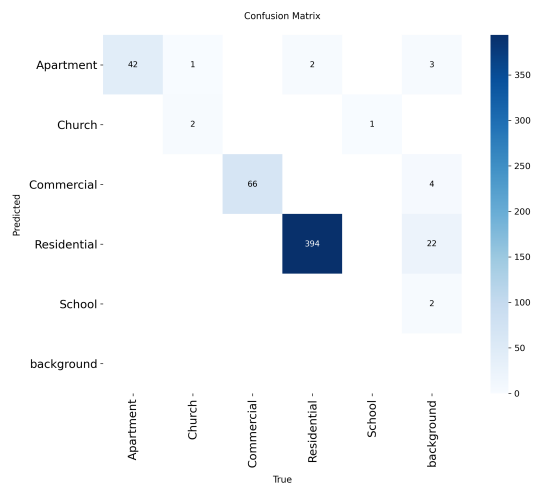
Research into other remote sensing data and how it could be incorporated.

V. ACKNOWLEDGMENT

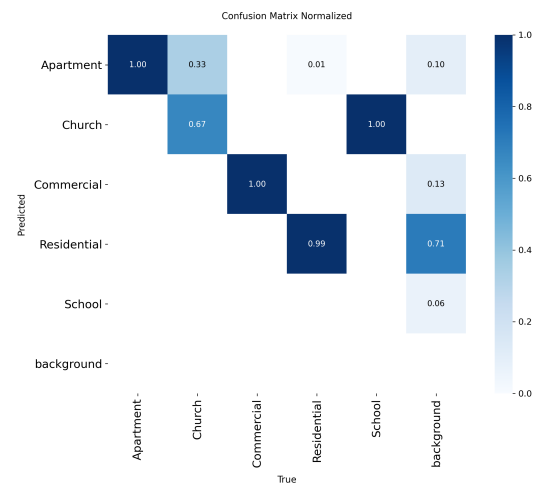
The authors would like to thank...

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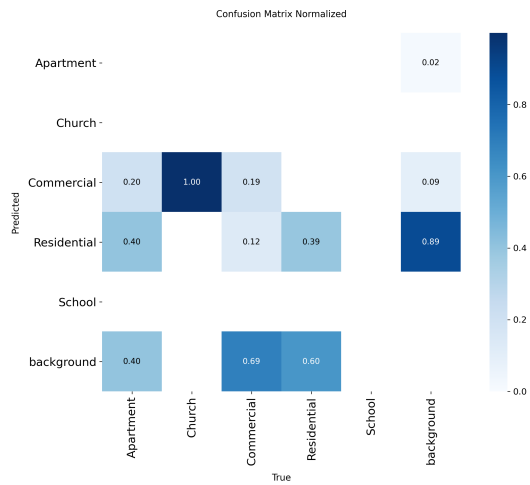
(a) Logan confusion matrix



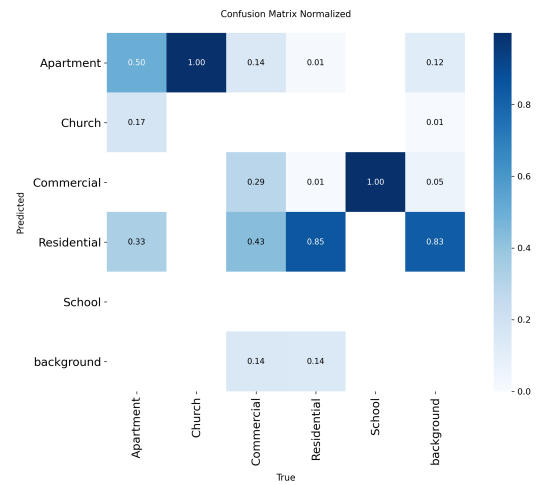
(b) Logan confusion Matrix normalized

Figure 3: YOLO confusion matrixes for Logan, Utah.

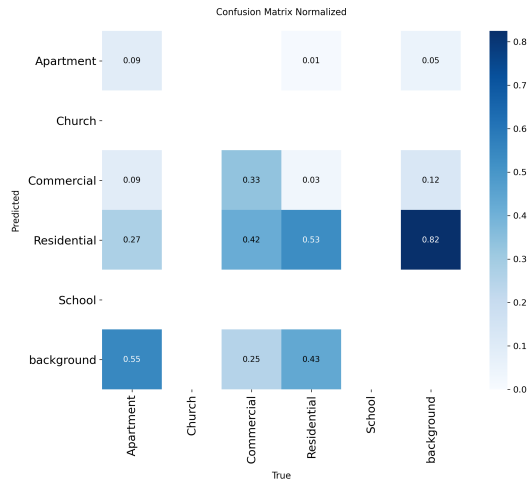
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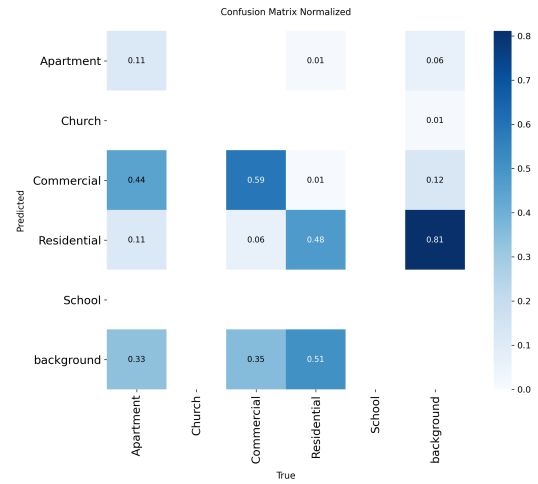
(a) Bountiful, Utah



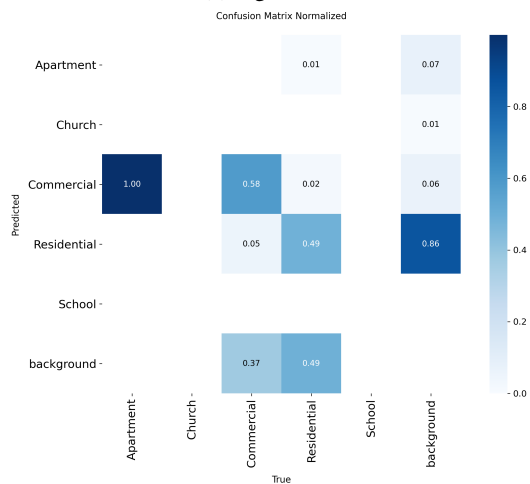
(b) Saratoga Springs, Utah



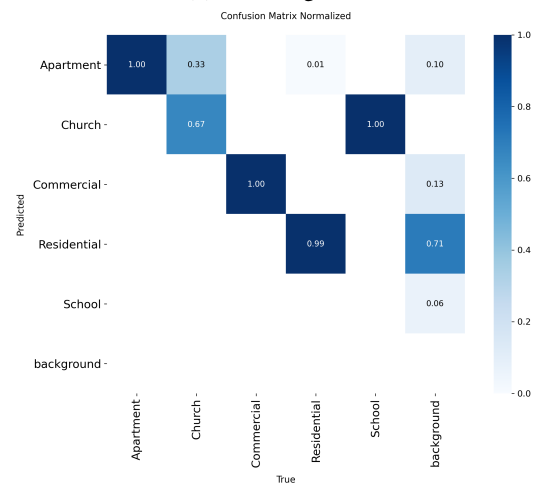
(c) Ogden, Utah



(d) St. George, Utah



(e) Price, Utah



(f) Logan Utah

Figure 4: Normalized YOLO confusion matrixes